

Petroller: Altering the Virtual Reality Controller with an Attachable Prop-based Haptic for Embodied Virtual Companion

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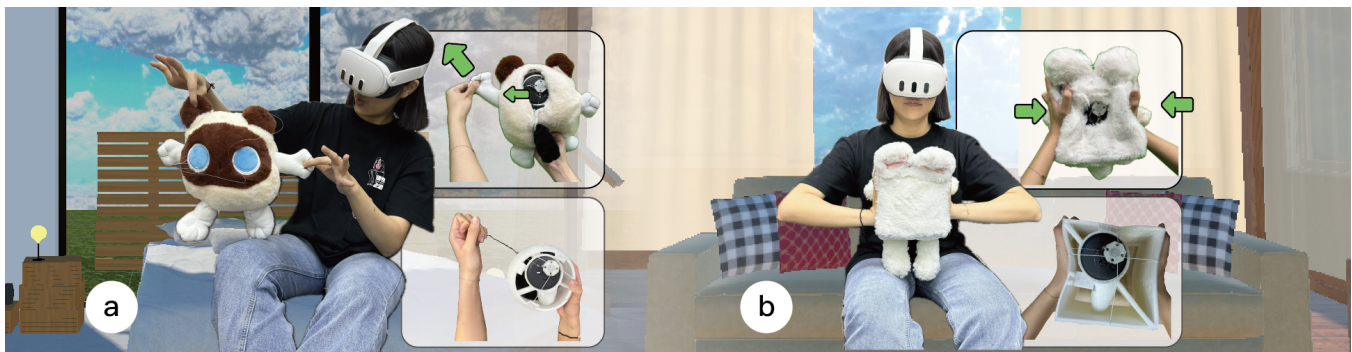


Figure 1: Petroller usage scenarios (a)pet companionship. (b)rehabilitation exercises.

Abstract

Previous studies have explored various forms of virtual companions, including displaying companions on flat screens, headsets, or the embodiment of their physical bodies. Although most studies use additional devices to enrich sensory feedback to embody the virtual companion in a virtual reality (VR) headset, it requires electricity to activate. We present “Petroller,” an attachable mechanism that embodies a virtual companion by altering the built-in VR controller with the prop-based haptic, allowing users to hug, pat, and exercise together. Additionally, we designed two usage scenarios: pet companionship and rehabilitation exercises. We aim for users to interact with the Petroller by pressing and stroking, providing relaxation, emotional healing, and even promoting physical exercise.

CCS Concepts

• **Human-centered computing** → **Interaction paradigms**; Virtual Reality.

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Keywords

Virtual Companion, Attachable, Prop-based Haptic, Virtual Reality

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1 INTRODUCTION

Previous research on Human-Animal Interaction has shown animal companionship significantly benefits various groups' psychological, social, and physical health. Touching or keeping animals as pets can positively impact human emotions [Wilson 1998]. Similarly, research like Bear-With-Me [Fong et al. 2013] has explored how tactile communication in emotionally interactive systems can deepen interpersonal connections, reinforcing the importance of touch in fostering emotional well-being. However, keeping a pet is not easy; it requires a significant investment of time and money, as well as learning how to care for the pet. Virtual pets can overcome these problems while still providing the joy of companionship. There are many ways to embody virtual pets; some do not involve physical objects and can only interact with the pets in the virtual world, such as mixed reality virtual pets accompanying overweight children during exercise [Johnsen et al. 2014]. Some provide physical objects

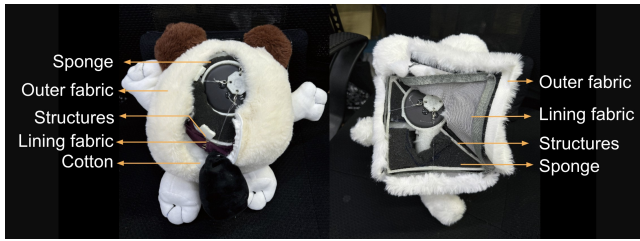


Figure 2: Petroller's mechanism.

to simulate pets and interact with children tangibly, such as the robotic dog AIBO [Stanton et al. 2008].

Many studies have incorporated devices such as trackers and evaluation boards to add additional functions. These devices provide haptic or sensory feedback through tangible objects, enhancing the sense of immersion. Pumping Life [Huang et al. 2019] utilizes a VIVE tracker and Arduino to detect pressure and control heating, providing users with tactile feedback. Real Baby - Real Family [Hsieh et al. 2017] uses two VIVE controllers to track and simulate a baby and a bottle for interaction. These devices offer users visual feedback in the virtual world and provide tactile sensations, enhancing the sense of immersion. Additionally, VRSurus [Du and He 2016] showcases how prop-based tactile feedback in VR, achieved through gesture recognition and smart puppetry, can further elevate the realism and engagement of virtual experiences. MMRPet [Xue et al. 2019] and [Dongye et al. 2023] modularize physical objects, allowing them to be assembled into various forms of pet companions, generating multiple types of haptic feedback. These studies explore the potential of combining virtual pets with physical objects. On the other hand, iturk [Cheng et al. 2018] provides additional feedback without requiring additional power systems to achieve tactile modifications. This has led us to propose an immersive virtual reality experience that simulates the tactile sensation of real pets. Users can feel the touch of pets while interacting with them in the virtual world, enhancing the sense of immersion.

2 DESIGN AND IMPLEMENTATION

Petroller is an immersive haptic feedback device designed to enhance the VR controller's user experience. This device is developed using Meta Quest 3 and Unity. We utilized 3D printing to create two shapes of attachable structures for the VR controller: a cube and a sphere. The structures are filled with sponge internally and wrapped externally with two layers of fabric. The inner layer connects parts such as limbs and tails using fishing lines, while the outer layer is covered with fabrics of varying lengths and materials. Velcro and zippers are used for fastening, allowing easy replacement of different skins to alter the tactile sensation of pet companions. To detect user-pressing actions, we designed cylindrical structures within the attachable structures that extend to the controller's trigger and grip buttons. These mechanisms transmit the user's pressing actions from the external structure to the internal controller, allowing the pressure to be effectively sensed. Additionally, we installed a circular structure on the joystick, connected to the external joints of the doll, to detect the tension when the user stretches the doll. These designs help users obtain realistic haptic feedback while interacting with virtual pets of different forms and hardness. In addition, the

buttons and joystick of the controller can detect the magnitude and direction of the user's applied force, allowing the system to determine which joint of the doll is being stretched or squeezed. The software then provides corresponding feedback to the virtual pet, such as changes in facial expressions or vocal responses, enhancing the sense of immersion.

3 APPLICATION: PETROLLER

To enable users to interact with virtual pets, we designed two scenarios to simulate using Petroller: (a) Virtual pet companionship: users can freely interact with Petroller in a bedroom setting by petting, rubbing, squeezing, and throwing. The user can even perform some usually impossible actions with real pets. (b) Home rehabilitation exercises simulate a situation where users perform rehabilitation exercises in their living room, accompanied by a virtual pet. We aim to provide relaxation and enhance the user's motivation to exercise through pet companionship.

4 DISCUSSION AND FUTURE WORK

In this work, we created an attachable mechanism that embodies a virtual companion by altering the built-in VR controller with haptic feedback but does not require electricity. Simulating two scenarios for Petroller enabled users to have enhanced interactions with their virtual pets. In the future, we will investigate the emotional relaxation and extent of relaxation experienced by users and whether it can enhance users' motivation to exercise.

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